



Year 12 Mathematics Extension 2
2013 HSC ASSESSMENT TASK 1
Term 4 Week 8 2012

Name: _____

Teacher: _____

Monday 26th November
Periods 5 & 6

Set by: VUL

- Attempt **all** questions.
- All questions are of equal value.
- Marks may be deducted for insufficient, or illegible work.
- Only Board approved calculators (**excluding** graphic calculators) may be used
- Total possible mark is **50**
- **Begin each question on a new page.**
- **TIME ALLOWED** : 90 minutes + 2 minutes reading time.

Question 1

Marks

- (a) Let $z = 2\sqrt{3} + i$ and $w = \sqrt{3} - i$
- (i) Find $2z - \bar{w}$ in the form $x + iy$ where x and y are real. **1**
- (ii) Find $\frac{z}{w}$ in the form $x + iy$ where x and y are real **2**
- (b) Find the two square roots of $3 - 4i$ for z giving your answers in the form $x + iy$ where x and y are real. **3**
- (c) (i) Express $-1 - \sqrt{3}i$ in modulus-argument form. **2**
- (ii) Hence evaluate $(-1 - \sqrt{3}i)^9$ **2**
- (d) On separate Argand diagrams, sketch the locus of points z such that:
- (i) $\arg(z - 1 + i) = \frac{\pi}{2}$ **2**
- (ii) the inequalities $|z - i| \leq 2$ and $1 \leq \text{Im}(z) \leq 2$ both hold **2**
- (iii) $|z + \bar{z}| = 1$ **2**
- (e) Find the three cube roots of $-8i$ in the form $x + iy$ where x and y are real. **3**

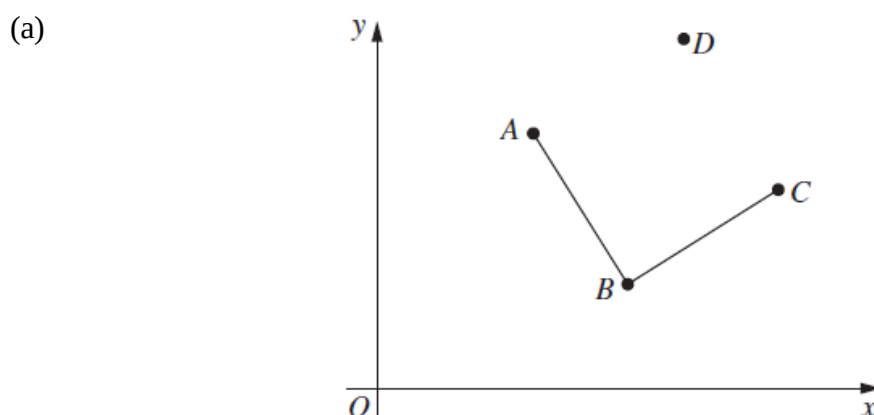
Question 1 is continued on the next page

Question 1 Continued

Marks

- (f) (i) By rationalising the numerator of $\frac{\sqrt{n+1}-\sqrt{n}}{1}$ prove that $\sqrt{n+1}-\sqrt{n} > \frac{1}{2\sqrt{n+1}}$. 2
- (ii) Hence prove by mathematical induction that $\sqrt{n} > 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ for $n \geq 7$. 4

Question 2 **Start a new page.**



In the diagram the vertices of a triangle ABC are represented by the complex numbers z_1 , z_2 and z_3 , respectively. The triangle is isosceles and right-angled at B .

- (i) Explain why $(z_1 - z_2)^2 = -(z_3 - z_2)^2$. 1
- (ii) Suppose D is the point such that $ABCD$ is a square. Find the complex number, expressed in terms of z_1 , z_2 and z_3 , that represents D . 2
- (b) (i) Sketch the locus of the complex number $z = x + iy$ where $\arg[z - 1] - \arg[z + 1] = \frac{\pi}{4}$. 1
- (ii) Find the Cartesian equation of the locus described in part (i). 1
- (iii) Give the range of the locus found in part (ii). 1

Question 2 is continued on the next page.

Question 2 Continued

(c) If $z = \cos \theta + i \sin \theta$:

(i) Show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$ 2

(ii) Use the binomial theorem to expand $\left(z - \frac{1}{z}\right)^5$ 1

(iii) Hence express $\sin^5 \theta$ in terms of $\sin n\theta$ 2

(d) (i) If ω is a complex root of $z^5 - 1 = 0$ with least positive argument, show that $\omega^2, \omega^3, \omega^4$ are the other complex roots. 2

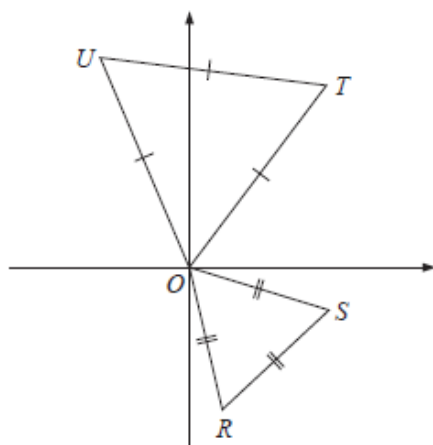
(ii) Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ 1

(iii) Plot all the roots of $z^5 - 1 = 0$ on an argand diagram. 1

(iv) Express $z^4 + z^3 + z^2 + z + 1$ as a product of two quadratic factors. 3

(v) Prove that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ 2

(e)



The diagram shows points O, R, S, T , and U in the complex plane. These points correspond to the complex numbers $0, r, s, t$, and u respectively. The triangles ORS and OTU are equilateral. Let $\omega = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

(i) Explain why $u = \omega t$. 1

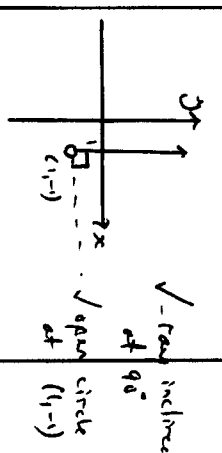
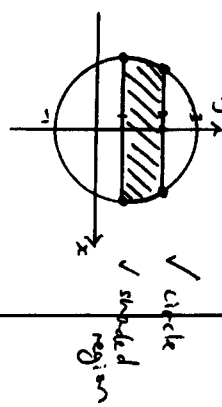
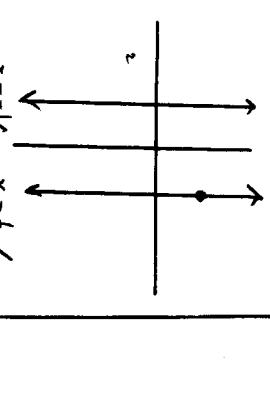
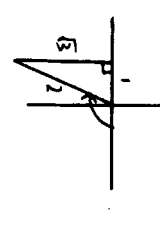
(ii) Find the complex number r in terms of s . 1

(iii) Using complex numbers, show that the lengths of RT and SU are equal. 3

End of Assessment Task



2013 Year 12 Mathematics Extension 2 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 1</u>			
a) i) $2z - \bar{z} = 3\sqrt{3} + i$	✓	d) i) $\arg(z - (1-i)) = \frac{\pi}{2}$	✓
ii) $\frac{z}{w} = \frac{25+i}{\sqrt{3}-i} \times \frac{\sqrt{3}+i}{\sqrt{3}+i}$	✓		✓ - ray inclined at 90° at open circle at (1, -1)
$= \frac{5}{4} + \frac{3\sqrt{3}}{4}i$	✓	ii) $ z-i \leq 2$ and $1 \leq y \leq 2$	✓
b) Let $(x+iy)^2 = 3-4i$	✓		✓ circle ✓ shaded region
$x^2 - y^2 + 2xyi = 3-4i$	✓ progress	iii) $ z + \bar{z} = 1$	✓
$x^2 - y^2 = 3$ $xy = -2$		$ 2x = 1$	✓
$x = \pm 2$ $y = \mp 1$		$x = \pm \frac{1}{2}$	✓
∴ square roots are $2-i$ and $-2+i$ (one for each root)			✓
c) i) 		ii) $(-1-\sqrt{3}i)^9 = [2\text{cis}(-\frac{3\pi}{2})]^9$	✓
$= -1 - \sqrt{3}i = 2\text{cis}(-\frac{3\pi}{2})$	✓	$= 2^9 \text{cis}(-6\pi)$	✓
ii) $(-1-\sqrt{3}i)^9 = [2\text{cis}(-\frac{3\pi}{2})]^9$	✓		
$= 2^9 \text{cis}(-6\pi)$	✓		
$= 2^9$ or 512	✓		



2013 Year 12 Mathematics Extension 2 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 1</u> (continued)			
e) Let $-8i = 8\text{cis}(-\frac{\pi}{2})$	✓	ii) when $n=7$	✓
$z^3 = 8\text{cis}(-\frac{\pi}{2})$	✓	$\sqrt{n} - (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n})$	
$z_k = 8^{\frac{1}{3}}\text{cis}(\frac{-\pi + 2k\pi}{3})$	✓	$= \sqrt{7} - (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{7})$	
for $k=0, 1, 2$	✓	$= \sqrt{7} - \frac{363}{140}$	
$z_0 = 2\text{cis}(-\frac{\pi}{6}) = \sqrt{3} - i$	✓	$\approx 0.0529 > 0$ from calc.	✓
$z_1 = 2\text{cis}(\frac{\pi}{6}) = 2i$	✓	Assume statement true for $m=k$	
$z_2 = 2\text{cis}(\frac{5\pi}{6}) = -\sqrt{3} - i$	✓	if $\sqrt{k} > 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$	
f) i) $\frac{\sqrt{n+1} - \sqrt{n}}{1} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$	✓	when $m=k+1$,	
$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$	✓	$\sqrt{n} - (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n})$	
$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$	✓	$= \sqrt{k+1} - (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} + \frac{1}{k+1})$	
$= \frac{1}{\sqrt{n+1} + \sqrt{n}} - \frac{1}{2\sqrt{n+1}}$	✓	$> \sqrt{k+1} - (\sqrt{k} + \frac{1}{k+1})$ by assumption	
$= \frac{\sqrt{n+1} + \sqrt{n} - 2\sqrt{n+1}}{2\sqrt{n+1}(\sqrt{n+1} + \sqrt{n})}$	✓	$= \sqrt{k+1} - \sqrt{k} - \frac{1}{k+1}$	
$\text{as } \sqrt{n+1} - \sqrt{n} > 0 \text{ and } (\sqrt{n+1} + \sqrt{n}) > 0$	✓	$> \frac{1}{2\sqrt{k+1}} - \frac{1}{k+1}$ from part i)	✓
$\therefore \sqrt{n+1} - \sqrt{n} > \frac{1}{2\sqrt{n+1}}$	✓	$= \frac{\sqrt{k+1}}{2\sqrt{k+1}} - \frac{1}{k+1}$ rationalising denominator	✓
		$= \frac{\sqrt{k+1} - 2}{2(k+1)} > 0$ for $k \geq 7$	✓
		∴ If statement true for $n=k$, then proved true for $n=k+1$, Since true for $n=7$, ∴ True for $n=8, 9, 10, \dots$	



2013 Year 12 Mathematics Extension 2 Task 1 SOLUTIONS

Suggested Solution (e)	Comments	Suggested Solution (f)	Comments
<u>Question 2</u> (a) i) $\vec{BA} = \hat{i} \times \vec{BC}$ anti-clockwise rotation of 90°, squaring we have $(z_1 - z_2)^2 = -(z_3 - z_2)$ as $\vec{BA} = z_1 - z_2$ $\vec{BC} = z_3 - z_2$ ii) $\vec{OD} = \vec{OA} + \vec{AD}$ $= \vec{OA} + \vec{BC}$ as $\vec{BC} = \vec{AD}$ opp. sides of square $\therefore \vec{OD} = z_1 + z_3 - z_2$ Alternative solutions at end of document b) i) ii) $z^2 + (y-1)^2 = 2$ iii) $0 < y \leq 1 + \sqrt{2}$	✓	c) $z^n = \cos n\theta + i \sin n\theta$ $z^n = \cos n\theta - i \sin n\theta$ $\therefore z^n - z^{-n} = 2i \sin n\theta$ ii) $(z - \frac{1}{z})^5$ $= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$ iii) When $m=1$, $(2i \sin \theta)^5 = z^5 - \frac{1}{z^5} - 5(z^3 - \frac{1}{z^3}) + 10(z - \frac{1}{z}) - \frac{10}{z^3} + \frac{5}{z^5}$ $32i^5 \sin^5 \theta = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta$ $\div 32i$ $\therefore \sin^5 \theta = \frac{1}{16} \sin 5\theta - \frac{5}{16} \sin 3\theta + \frac{5}{8} \sin \theta$ d) i) The roots of $z^5 - 1 = 0$ are equally spaced $\frac{2\pi}{5}$ radians apart on the unit circle starting with $z_0 = 1$ So let $z_1 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} = w$ $\therefore z_2 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} = w^2$ $z_3 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} = w^3$ $z_4 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} = w^4$ ii) $1 + w + w^2 + w^3 + w^4$ $= \text{sum of roots}$ $= -\frac{-1}{1} \text{ as } z^5 - 1 = 0$ $= -\frac{0}{1} = 0$	✓



2013 Year 12 Mathematics Extension 2 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
iii) iv) $z^5 - 1$ $= (z - w)(z - w^2)(z - w^3)(z - w^4)(z - 1)$ $= (z - w)(z - w^2)(z - w^3)(z - w^4)(z - 1)$ $= [z^2 - 2\operatorname{Re}(w)z + w ^2][z^2 - 2\operatorname{Re}(w^2)z + w^2 ^2](z - 1)$ $= (z^2 - 1)(z^2 - 2\cos\frac{2\pi}{5}z + 1)(z^2 - 2\cos\frac{4\pi}{5}z + 1)$ ✓ ✓ (One for each quadratic factor) Since $z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$ $\therefore z^4 + z^3 + z^2 + z + 1$ $= (z^2 - 2\cos\frac{2\pi}{5}z + 1)(z^2 - 2\cos\frac{4\pi}{5}z + 1)$	✓	v) w and w^4 are roots of $z^2 - 2\cos\frac{2\pi}{5}z + 1$ $\therefore w + w^4 = \frac{-b}{a} = 2\cos\frac{2\pi}{5}$ — ① Similarly w^2 and w^3 are roots of $z^2 - 2\cos\frac{4\pi}{5}z + 1$ $\therefore w^2 + w^3 = \frac{-b}{a} = 2\cos\frac{4\pi}{5}$ — ② ✓ for progress in ①+②: $w + w^2 + w^3 + w^4 = 2(\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5})$ but $1 + w + w^2 + w^3 + w^4 = 0$ $\therefore -1 = 2(\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5})$ $\therefore \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$ c) Since ΔTOS is equilibrated $\angle TOU = \frac{\pi}{3}$ Now $\vec{OU} = \vec{OT} \times \cos\frac{\pi}{3}$ (vector in direction of $\vec{T}$) $\therefore u = w^4$ ii) Similarly $\vec{OR} = \vec{OS} \times \cos(-\frac{\pi}{5})$ (clockwise rotation of $-\frac{\pi}{5}$) $\therefore r = \bar{w}^5$ or equivalent ✓ $= \frac{5}{w}$ $r = 1/5 \operatorname{cis}(-\frac{\pi}{5} + \arg s)$	✓ verified progress



2013 Year 12 Mathematics Extension 2 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 2</u> Continued. iii) Since $\vec{SA} = u - s$ and $\vec{AT} = t - r$ $u \times \vec{AT} = u \times t - u \times r$ $= u \times s$ as if $r = \bar{u}s$ $s = ur$ Now $u \times \vec{AT} = u - s$ $\therefore u \times \vec{AT} = u - s$ but $u = 1$ $\therefore x/rt = u - s $ $ rt = u - s $ $= sv $ as required * See attached document for alternate solutions	✓ ✓ ✓ ✓		



2013 Year 12 Mathematics Extension 2 Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
ALTERNATIVE SOLUTIONS to QUESTION 2(a) (ii) Let $D(z_4)$. <u>VERSION 1</u> $\vec{CD} = -i \vec{CB}$ $z_4 - z_3 = -i(z_2 - z_3)$ $z_4 = z_3 + i(z_3 - z_2)$ $z_4 = (1+i)z_3 - i z_2$ <u>VERSION 2</u> $\vec{BD} = \vec{BC} + \vec{BA} \quad (\text{parallelogram vectors})$ $= (z_3 - z_2) + (z_1 - z_2)$ i.e. $\vec{BD} = z_1 + z_3 - 2z_2$ AND $\vec{BD} = \vec{OD} - \vec{OB}$ So $z_4 - z_2 = z_1 + z_3 - 2z_2$ $\Rightarrow z_4 = z_1 + z_3 - z_2$ <u>VERSION 3</u> $\vec{AD} = i \vec{AB}$ $= i(z_2 - z_1)$ i.e. $z_4 - z_1 = i(z_2 - z_1)$ $\Rightarrow z_4 = z_1 + i(z_2 - z_1)$ $z_4 = (1-i)z_1 + i z_2$		<u>VERSION 4</u> $\vec{OD} = \vec{OC} \times \sqrt{2} \times \text{cis } \frac{\pi}{4}$ dilation factor $= (z_3 - z_2) \times \sqrt{2} \times \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right)$ $= (1+i)(z_3 - z_2)$ or $z_4 = (z_3 - z_2) + i(z_3 - z_2)$ rotation factor <u>VERSION 5</u> $\vec{BD} = i \vec{AC}$ $z_4 - z_2 = i(z_3 - z_1)$ $\Rightarrow z_4 = z_2 + i(z_3 - z_1)$ or $z_4 = z_2 + i z_3 - i z_1$ <u>VERSION 6</u> $\vec{AD} = i \vec{AB} \Rightarrow z_4 - z_1 = i(z_2 - z_1) \quad \text{--- (i)}$ & $\vec{CD} = i \vec{BC} \Rightarrow z_4 - z_3 = i(z_3 - z_2) \quad \text{--- (ii)}$ (i) + (ii) gives: $2z_4 - z_1 - z_3 = i(z_3 - z_1)$ $\Rightarrow 2z_4 = (z_1 + z_3) + i(z_3 - z_1)$ $\Rightarrow z_4 = \frac{1}{2}(1-i)z_1 + \frac{1}{2}(1+i)z_3$ or $z_4 = \frac{1}{2}(z_1 + z_3) + \frac{1}{2}i(z_3 - z_1)$	